

CASCADE: an open-source Python package for onboard Bayesian triage of agricultural drought from Earth observations

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Abstract

CASCADE is an open-source Python package for evaluating onboard Earth-observation triage policies for agricultural drought and crop-stress monitoring. It combines a Crop Stress Composite (CSC) from vegetation, thermal, and moisture anomalies with a Bayesian belief gate and finite-horizon scheduling policy. Each pass is skipped, screened, fused, or downlinked as a priority alert. The package includes deterministic simulation, MODIS replay, calibration, validation fixtures, and paper-grade artifacts. In a 100-seed Monte Carlo benchmark, CASCADE reduces downlink by 99.05% while preserving 100.0% anomaly-event recall at 0.6% false positives.

Keywords: onboard autonomy, Earth observation, drought monitoring, Bayesian inference, satellite scheduling, remote sensing

Required Metadata

Current code version

1. Motivation and significance

Earth-observation missions increasingly collect more imagery than they can downlink or inspect immediately. Agricultural drought monitoring is a sharp example: growers, crop advisors, and food-security analysts need early warnings for specific field polygons, while conventional workflows often downlink broad scenes and wait for ground processing [1]. This creates a mismatch between the spatial precision of modern farm decisions and the bandwidth, energy, and latency limits of small satellite systems.

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Nr.	Code metadata description	CASCADE value
C1	Current code version	v1.0.0
C2	Permanent GitHub link to code/repository used for this code version	https://github.com/emillambert/CASCADE/releases/tag/v1.0.0
C3	Legal Code License	MIT
C4	Code versioning system used	git
C5	Software code languages, tools, and services used	Python 3.10+; requests; numpy; pandas; matplotlib; rasterio; tqdm; pytest; GitHub Actions
C6	Compilation requirements, operating environments & dependencies	Tested on Linux and macOS; Python ≥ 3.10 , < 3.13 ; install with <code>pip install .</code> or <code>pip install cascade-eo</code> ; dependencies in <code>pyproject.toml</code>
C7	If available Link to developer documentation/manual	https://github.com/emillambert/CASCADE/tree/main/docs
C8	Support email for questions	e.w.lambert@student.tudelft.nl; issue tracker: https://github.com/emillambert/CASCADE/issues

Table 1: Code metadata (mandatory).

CASCADE addresses this gap as research software for testing onboard triage policies. Instead of treating a satellite pass as a fixed acquisition, CASCADE asks whether the current field state justifies additional sensing and downlink. The software frames crop-stress monitoring as a scheduling problem in which each pass selects one of four actions: skip, perform vegetation screening, fuse multiple stress channels, or downlink a priority alert. The package therefore supports research on how much autonomy can be moved onboard before flight hardware is available.

The scientific problem is not drought classification alone. It is the joint problem of stress detection, posterior belief management, and resource-aware sensing under constrained spacecraft budgets. CASCADE combines MODIS-derived vegetation, land-surface-temperature, and reflectance products [2, 3, 4] into an anomaly score while retaining the traceability expected for early flight-software verification. This formulation draws on anomaly detection [5], Bayesian updating [6], and finite-horizon Markov decision processes [7]. It distinguishes CASCADE from purely ground-based remote-sensing pipelines and from general mission-planning systems such as ASPEN and EO-1 autonomous science operations [8, 9, 10].

CASCADE also occupies a different niche from general geospatial and

optimization packages. Libraries such as raster processing tools, array frameworks, Bayesian toolkits, or POMDP solvers provide useful primitives, but they do not package the full experiment: CSC construction, a Beta-posterior gate, resource-aware action selection, offline MODIS replay artifacts, and CI-backed fixtures for the same drought-triage benchmark. CASCADE is therefore intended as a small, auditable reference implementation rather than a general-purpose scheduler.

This manuscript formalizes CASCADE as the citable software artifact for the scheduler, synthetic benchmark, and real-scene Westlands replay. Users can run the benchmark, replay the accepted MODIS evidence, inspect calibrated parameters, and modify the policy thresholds or action budgets. This makes CASCADE useful both as a domain-specific drought triage tool and as a compact testbed for onboard autonomy research in other anomaly-monitoring domains.

2. Software description

2.1. Software architecture

CASCADE is organized as a Python package under `src/cascade/`. The core module defines the hardware budget, action table, Crop Stress Composite (CSC), Bayesian posterior helpers, and deterministic scheduling policy. The simulation module generates synthetic seasons, evaluates raw, fixed-period, EVI-only, no-belief, and full CASCADE baselines, and writes benchmark artifacts. The replay module wraps AppEEARS/MODIS processing and also exposes offline artifact-backed replays for review. Calibration and economics modules generate additional release artifacts used by the paper and validation tests.

The repository separates transient and release-grade outputs. Generators write to `build/`; reviewed outputs are promoted into `artifacts/`; tests compare accepted JSON fixtures against those tracked artifacts. This layout lets reviewers reproduce headline claims without live credentials while preserving a live AppEEARS path for users who need to replay new areas.

2.2. Software functionalities

The primary public functions live in `cascade.core` and `cascade.replay`. The CSC maps vegetation loss, thermal rise, and moisture loss into a bounded stress index. For patch i ,

$$\text{CSC}_i = \text{clip} \left(w_E \frac{[E_{0,i} - E_i]_+}{s_E \sigma_E} + w_L \frac{[L_i - L_{0,i}]_+}{s_L \sigma_L} + w_N \frac{[N_{0,i} - N_i]_+}{s_N \sigma_N}, 0, 1 \right),$$

CASCADE software modules and data flow

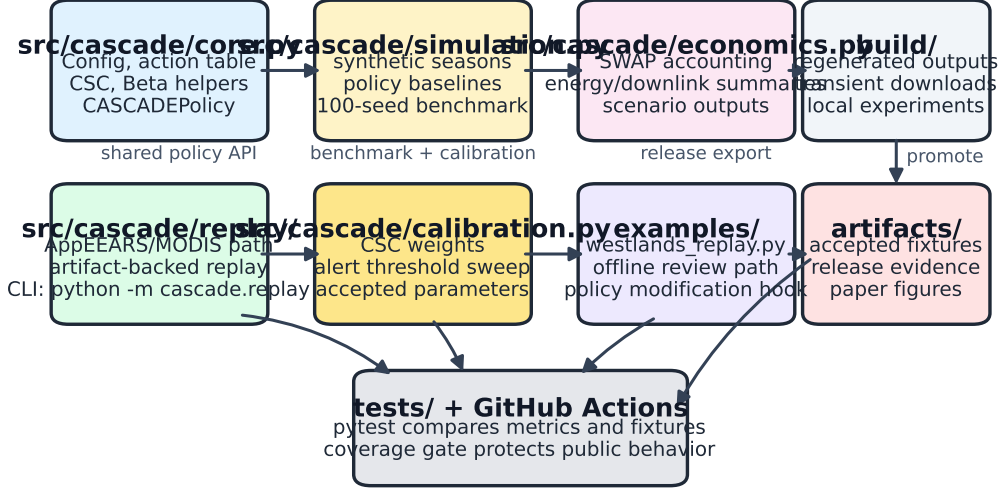


Figure 1: CASCADE software architecture. Package modules share the public policy and replay APIs, write transient outputs to `build/`, promote accepted evidence to `artifacts/`, and are checked by tests and GitHub Actions.

where E , L , and N denote EVI, land-surface temperature, and NDWI. The MOD13 action is the vegetation-only screening pass based on the MOD13A1 product.

The belief update stores patch-level evidence in Beta posterior parameters. A positive screening observation increments α_i ; otherwise β_i is incremented, with bounded evidence rescaling:

$$(\alpha_i, \beta_i) \leftarrow (\alpha_i + \mathbf{1}_{z_i}, \beta_i + \mathbf{1}_{\neg z_i}), \quad p_i \sim \text{Beta}(\alpha_i, \beta_i).$$

`CASCADEPolicy.act` then chooses between `SKIP`, `MOD13`, `FUSE`, and `FUSE_PRIORITY`-eligible fusion based on state of charge, posterior tail probability $\Pr(p_i > 0.5)$, and time since last fusion. The default CSC alert threshold, 0.615, is the calibrated operating point from the synthetic threshold sweep and can be changed for sensitivity analysis.

Command-line entry points support the same workflows. The simulation command runs the fast benchmark path, while the replay command selects either the 2014 or 2024 Westlands evidence and prefers tracked artifacts when present. Calibration and economics scripts regenerate the matching release artifacts. The SoftwareX example, `examples/westlands_replay.py`, is intentionally short and offline by default.

3. Illustrative examples

The main worked example is the 2014 Westlands Water District (Firebaugh, California) D4 drought replay, referred to below as Westlands. From a clean checkout, a user installs the package and runs:

```
python examples/westlands_replay.py
```

The script calls the replay API, reads the tracked `replay_metrics.json` artifact, and prints the peak CSC, number of `FUSE_PRIORITY` windows, action distribution, and whether the source was tracked artifacts or a live replay. No Earthdata credentials are needed for the default path.

```
CASCADE Westlands replay (2014)
```

```
source: artifacts
```

```
peak CSC: 0.869
```

```
FUSE_PRIORITY windows: 6
```

```
action distribution:
```

```
  SKIP: 0
```

```
  MOD13: 0
```

```
  FUSE: 0
```

```
  FUSE_PRIORITY: 6
```

For live operation, users can run `real_modis_replay.py` with Earthdata credentials and an AppEEARS-accessible area of interest. To test another agricultural region, they can pass a new bounding box or area label through the replay command and keep the same policy thresholds. To test another on-board policy, they can subclass or replace `CASCADEPolicy`, then compare the regenerated action distribution, downlink volume, and alert windows against the accepted fixtures. The live path writes fresh outputs to `build/replay/`, keeping externally dependent downloads separate from the default offline review workflow.

4. Impact

CASCADE enables new research questions about onboard autonomy for Earth observation: how much posterior evidence is enough to justify high-resolution confirmation, how energy and downlink constraints change optimal sensing cadence, and how a policy calibrated only on synthetic data behaves on historical real scenes. Because the action set and posterior gate are anomaly-index agnostic, the same framework can be tested for wildfire

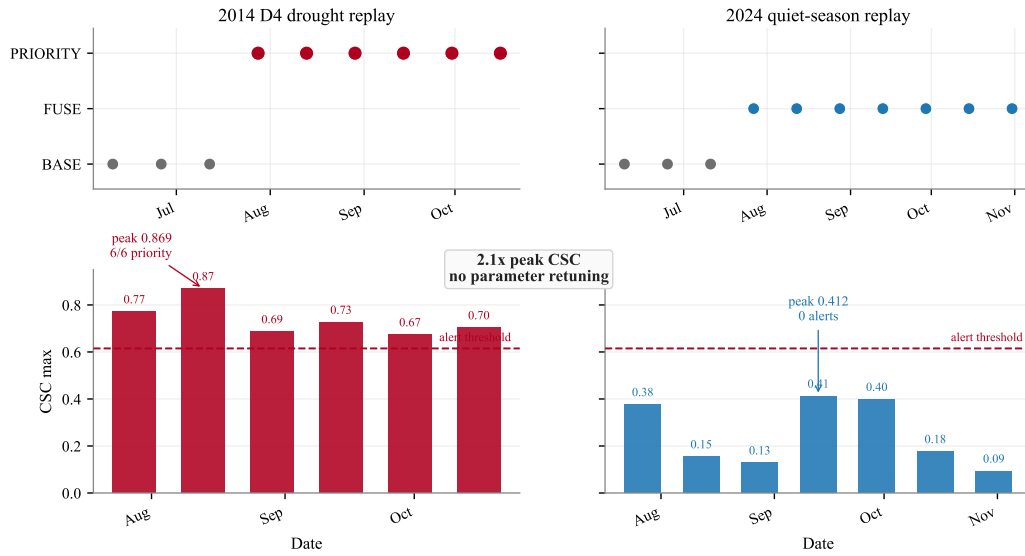


Figure 2: Westlands replay with unchanged policy parameters. MOD13 denotes vegetation-only screening, FUSE denotes full CSC generation, and FUSE_PRIORITY denotes fused alert tiles during downlink windows. The 2014 D4 drought season triggers priority alerts in every active post-warmup window, while the 2024 quiet season remains below the calibrated CSC alert threshold of 0.615.

(a) Peak CSC field, Westlands 2014

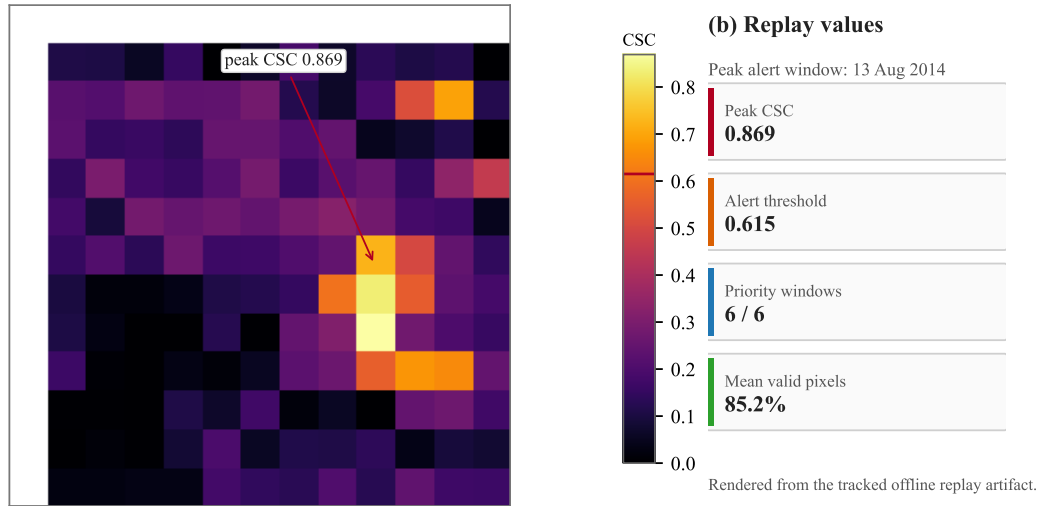


Figure 3: Peak 2014 alert map generated from the tracked replay artifact. White cells indicate invalid or masked MODIS pixels; the side panel reports the replay values exposed by the example script and validation fixtures.

fuel moisture, flood-front tracking, or urban heat monitoring by substituting the stress index while keeping the scheduling interface unchanged.

The package also improves existing remote-sensing questions. Many drought and crop-stress studies assume that imagery has already reached the ground. CASCADE lets researchers include the spacecraft decision layer in the experiment. The strongest concrete result is the 100-seed Monte Carlo benchmark: full CASCADE reduces downlink by 99.05% versus raw collection, saves 38.3% energy, preserves 100.0% anomaly-event recall, and holds a 0.6% false-positive rate. Against a fixed onboard baseline it reduces downlink by 77.6% and energy by 25.4%. These values are protected by validation tests and accepted fixtures, so changes to the scheduler or CSC calibration must either preserve them or update the evidence trail.

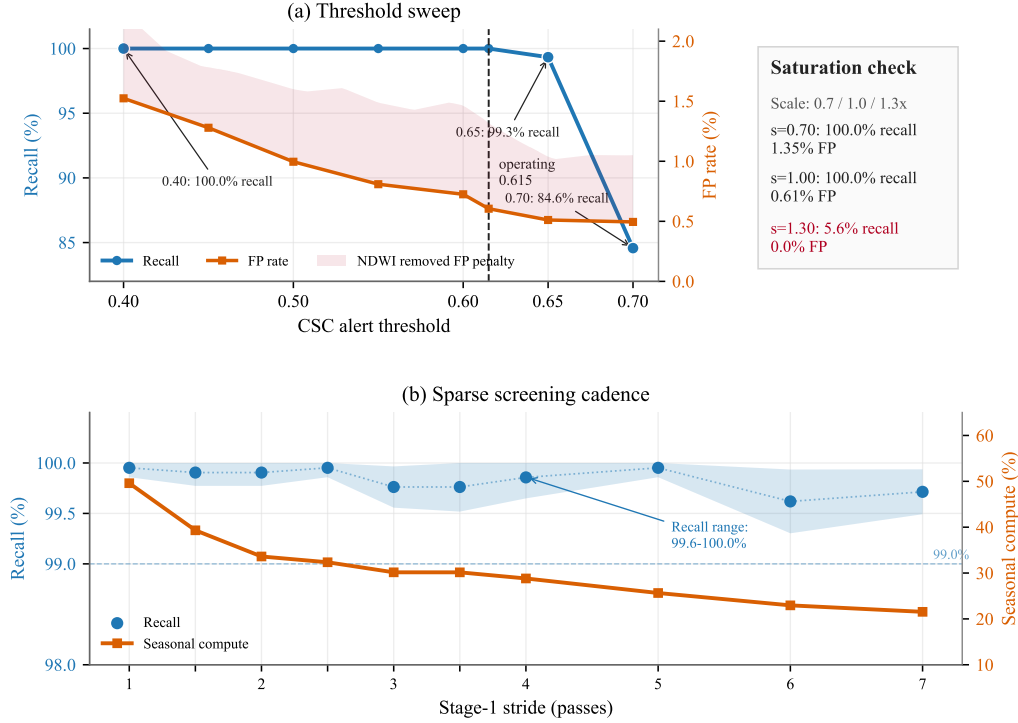


Figure 4: Benchmark ablations. Threshold and cadence sweeps show how alert sensitivity, false positives, and compute load move around the calibrated CSC operating point of 0.615. The plotted operating point is the threshold used by the replay examples and validation fixtures.

The Westlands replay demonstrates how the same committed policy transfers to archival real scenes. The 2014 California exceptional drought replay reaches peak CSC 0.869 and issues six priority alerts across six active post-warmup windows. The 2024 quiet-season replay peaks at CSC 0.412

and issues no priority alerts. Both runs use the same calibrated 0.615 alert threshold and the same tracked code path, providing a reproducible bridge between synthetic calibration and historical evidence [11, 12].

Daily practice impact is currently prospective because CASCADE is pre-flight research software, not an operational advisory service. Its near-term use is by researchers and mission designers who need to compare policy choices before hardware-in-the-loop or hosted-payload testing. The package changes that workflow by replacing spreadsheet-level resource assumptions with executable replay, fixtures, and CI. Uptake at submission time is therefore measured by open repository availability, tests, artifacts, and reproducible release materials rather than downloads, deployed users, or realized commercial revenue.

5. Conclusions

CASCADE provides an executable, tested package for studying onboard Bayesian triage of crop stress from Earth observations. Its contribution is the integration of a CSC anomaly score, Beta-posterior belief gate, resource-aware action policy, simulation benchmark, and MODIS replay harness in a single open-source repository. The current release reproduces the Westlands drought/quiet-season separation and protects paper-facing metrics through CI and accepted artifacts. Future work will extend the replay set beyond Westlands, run hardware-in-the-loop tests on flight-like processors, and generalize the anomaly-index interface to additional environmental monitoring domains.

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Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The source code, validation fixtures, and release artifacts are available at <https://github.com/emillambert/CASCADE>. The SoftwareX code metadata table points to the intended v1.0.0 GitHub release. Live MODIS replay

depends on NASA AppEEARS and Earthdata credentials; offline replay artifacts needed for the illustrative examples are included in the repository.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During manuscript preparation, the author used OpenAI Codex for planning, repository inspection, wording suggestions, and LaTeX drafting assistance. The author reviewed and edited the manuscript, verified the software outputs, and remains responsible for all scientific claims, citations, and submission materials. No scientific data or benchmark results were generated by AI tools.

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Current executable software version

CASCADE is a pure-Python package. The executable software version is the same as the source-code release listed in Table 1: v1.0.0 at <https://github.com/emillambert/CASCADE/releases/tag/v1.0.0>. There is no separate compiled binary artifact.